

G1 Digital-Twin Payload Experiment

Methods, spill model, governance algorithm, full campaign results and theory-consistency audit — supporting material for Section VII.H

Simulation campaigns 2026-07-04 · Unitree G1 · Gazebo (ROS 2 Humble) · 103 simulation runs, 86 valid governed/baseline runs across 12 arms · rev. 3 (adds pre-deployment validation of the four-layer configuration, live $\hat{p}$ traces, and the claim-by-claim consistency audit)

1. Digital twin: Gazebo replica of the laboratory

The physical robot is a Unitree G1 “Air” humanoid, a consumer unit whose only external interface is a single WebRTC session through the vendor app; the production navigation stack drives it by tapping that session over USB. For the twin we reuse **the exact same stack, unmodified**, and substitute only its interface object, so any behaviour observed in simulation is produced by the identical code deployed on hardware.

The simulator runs in a Docker container (ROS 2 Humble Desktop, Gazebo Classic, noVNC). The world, **lab.world**, is generated programmatically from the robot's own SLAM maps of the real laboratory: wall occupancy cells become 2.2 m box obstacles in the robot's coordinate frame, so the twin's geometry, waypoints (A, B, C) and the 0.8 m doorway coincide with the real deployment.

1.1 Robot model and physics

The robot is spawned from a URDF whose collision geometry is a single box matching the G1's real envelope (0.44 m shoulder width × 1.10 m tall, capped below the simulated LiDAR at $z \approx 1.20$ m so the sensor does not see itself). High mass (60 kg) and large diagonal inertia keep the base upright under the velocity-controlled gait; ground friction is zero so the planar-move plugin drives it as a velocity-controlled puck while wall contacts remain solid. Locomotion is commanded as gait velocity through `gazebo_ros_planar_move (/cmd_vel, /odom back)`; a 360-ray LiDAR at 10 Hz publishes `/scan`. The doorway was re-carved to the measured physical opening after the collision model was corrected: with the original carve the clear gap was 0.42 m and the realistic body physically could not pass — an artefact caught by consistency checking before any campaign data was taken.

1.2 The adapter: same stack, simulated interface

- **Velocity commands** $\{l_x, l_y, r_x, r_y\} \rightarrow /cmd_vel$, using the physical robot's measured stick-to-velocity mapping (deadzone subtracted, not clipped; $l_y = 0.40 \rightarrow 0.30$ m/s), so simulated and real dynamics coincide.
- **Pose** $\leftarrow /odom$ (map and odometry frames coincide in simulation: no relocalisation jitter).
- **LiDAR** $\leftarrow /scan$, projected into a flat world-frame point cloud identical in shape to the real WebView decoder output.

Camera perception is disabled in the twin; the LiDAR-driven planner and the governance layer are exercised in full. Data runs are always headless.

2. Spill model (payload hazard)

Spilling is modelled physically, not as an arbitrary speed threshold. The water surface at the cup rim is a **damped second-order oscillator** — the first sloshing mode of an open cylinder — excited by the horizontal acceleration of the cup:

$$\omega_0 = \sqrt{g \cdot k_1 \cdot \tanh(k_1 \cdot h)}, \quad k_1 = 1.841 / R \quad (\approx 21 \text{ rad/s for } R = 4 \text{ cm, } h = 8 \text{ cm})$$
$$\ddot{\eta} = -2 \cdot \zeta \cdot \omega_0 \cdot \dot{\eta} - \omega_0^2 \cdot (\eta + R \cdot a_{\text{cup}}/g) \quad \text{per body axis, } \zeta = 0.06$$

The cup is held ~0.25 m ahead of the base, so $a_{\text{cup}} = a_{\text{base}} + \alpha \times r + \omega \times (\omega \times r)$: fast turning adds a lever term. Bipedal gait injects acceleration noise growing with speed ($\approx v^2$; 1.4 Hz component plus broadband) — **walking fast agitates the cup by itself**. A sharp deceleration while forward motion is still commanded (an external impact, not a commanded stop) adds a direct impulse. Commanded acceleration is low-pass filtered ($\tau = 0.25$ s) because the real G1 ramps velocity smoothly whereas the planar-move plugin steps it instantaneously.

Spills are drawn from a **non-homogeneous Poisson process** on how far the surface exceeds the freeboard (12 mm below the rim):

$$\lambda(t) = \text{LID} \cdot \lambda_0 \cdot \max(0, |\eta|/\text{freeboard} - 1)^2, \quad \lambda_0 = 40 \text{ s}^{-1}, \quad \text{LID} = 1.0 \text{ (open)} / 0.25 \text{ (sealed)}$$

Each spill loses water (freeboard grows) and dissipates slosh energy. The model is **observational only**: it never alters physics or control. Calibration against logged trajectories yields ≈ 2 spills per 80 s of sustained 0.30 m/s cruising and ≈ 0 at the governed 0.18 m/s ceiling, consistent with the jerk-dominated design of the 2-D thesis study. For the physical robot the same event channel accepts a **manual ground-truth mark** (a keypress or an empty ROS message per observed spill), timestamped into the identical dataset field.

3. Governance algorithm (DCE runtime)

The governance layer is the configuration-first meta-reasoner (Meta-Reasoner 2.0) fed at 2 Hz through a bridge that packages the stack's per-tick telemetry as shared-experience readings. Each tick the reasoner evaluates the candidate analogies (Efficient_Nav / Cautious_Nav) against their QoE boundaries — analogy-level DST intervals, semantic-region memory and direction, tension with attention exponents, fulfillment, required-meta, uncertainty and hard-veto gates — and returns KEEP / SWITCH / FALLBACK / HELP / INSUFFICIENT plus the active analogy. The bridge maps the decision to a forward-speed ceiling (Efficient = uncapped ≈ 0.30 m/s, Cautious = 0.18 m/s, FALLBACK = 0.14 m/s, HELP = 0).

Meta-parameter	Source (per-tick telemetry)	Role
safety	clearance = $c_0 / 1.5$ m (LiDAR forward free space)	QoE regions + hard veto
progression	goal-approach rate / 0.30 m/s (2 s window)	required-meta threshold
mobility	achieved / commanded speed (1.2 s window)	resistance channel, hard veto
fragility	payload state: 1.0 intact, drops per spill	hard veto (payload task)
reliability / uncertainty	sensing monitor + few-shot historical margin	DST belief interval

Readings are median-smoothed (~ 2.5 s) and each reading's DST uncertainty adds the historical dispersion of its own metric (few-shot margin). Switches are hysteretic; an experience-escalation layer aborts on sustained HELP or a stagnant high-FALLBACK window.

3.1 Realising the four-layer belief architecture of thesis Chapter 5

Chapter 5 of the thesis specifies three DST belief layers plus a deployment-level Meta-Attention layer, each addressing a distinct failure mode at its own timescale. Table 2 maps each layer to its realisation on the G1. Layers 1-2 ran in every governed campaign arm; Layers 3-4 are opt-in (disabled during the reported campaign so the published arms remain exactly reproducible) and were validated in the loop by the pre-deployment arm of §5.1.

Thesis layer	Timescale	G1 realisation	Status
L1 — path-hazard plausibility (PI_H): collapse transfer, force conservative	sensor frame	Per-tick analogy-level DST on the safety reading with QoE regions and hard veto inside the reasoner (strips certification in ~ 2 ticks; measured ≈ 4 s correction of a task-blind prior), over the stack's own hard safety guards	covered (equivalent mechanism)
L2 — analogy transfer trust (PI_analogy): outcome evidence per run	run	Per-analogy Dempster–Shafer belief (match, mismatch, $\emptyset$) persisted between runs (G1_M2_STATE); spill $\rightarrow$ mismatch mass, clean run $\rightarrow$ match mass; $PI < 0.70$ blends the ceiling toward conservative, $PI < 0.45$ vetoes deployment and corrects the start-up prior	implemented; exercised in campaign
L3 — environment relevance (PI_env): relax / escalate spatial assumptions	run (continuous)	EMA of the median clearance relative to the active analogy's expected boundary, clamped to [0.85, 1.15]; scales profile ceilings only (never FALLBACK/HELP; bounds 0.22-0.40)	implemented; in-loop validated (§5.1)
L4 — Meta-Attention: chronic low fulfilment despite high trust $\rightarrow$ profile mismatch	deployment (run window)	Per-run mean fulfilment history persisted with the L2 state; fulfilment < 0.45 over 3 consecutive runs with $PI \geq 0.45$ under the same profile $\rightarrow$ start-up re-selection of the alternative profile. The experience-escalation abort covers the within-run part of this timescale	implemented; in-loop validated (§5.1)

Table 2. Thesis Ch. 5 four-layer architecture mapped to the G1 runtime. All four layers live in the bridge; the meta-reasoner itself is untouched.

Two details differ from the 2-D study, both additive. First, the **fragility channel acts within the run** (one spill $\rightarrow$ caution ≈ 0.52 , two $\rightarrow$ FALLBACK band ≈ 0.31 , three or more $\rightarrow$ dangerous < 0.12 and a hard-veto HELP; ~ 25 s recovery), which the 2-D study could not do because its spill evidence arrived only per run. Second, spill evidence is attributed to the analogy **governing at the moment of the spill**, sharpening the Layer-2 credit assignment.

3.2 Runtime certification ratio (ρ_{DCA})

The paper's robustness certificate is instantiated as a runtime quantity and logged with every governance decision:

$$\hat{\rho} = \max(\theta, \Delta\text{margin}) / (\text{task_uncertainty_gap}(\text{winner}) + 0.5 \cdot \theta_{\text{mismatch}}(\text{winner}) + 0.02)$$

where Δmargin is the stable-fulfilment gap between the two best deployable candidates (or the distance to the certification threshold when only one is deployable), $\text{task_uncertainty_gap}$ is the DST-measured calibration budget (the θ_{QoE} proxy), and θ_{mismatch} is the L2 distance between normalised task and analogy attention vectors (B constants set to 1; ξ_{rep} declared unmeasured at runtime). Controlled validation reproduces the paper's regimes: $\hat{\rho} = 1.50$ with clean readings versus $\hat{\rho} = 0.58$ under degraded, oscillating readings. Live traces are analysed in §5.1.

4. Campaign protocol

103 simulation runs were executed in total: 82 valid formal-campaign runs across 11 governance arms (8 per arm; 6 for sealed-lid arms), a 4-run pre-deployment validation arm (§5.1), plus development and validation runs, and one run excluded by audit (governance silently disabled by a mid-campaign code edit — detected by a per-tick telemetry check and discarded). Every run resets Gazebo to a fresh world with the robot at waypoint A and executes an autonomous A→B traverse headless; arms alternate to share machine-load drift; datasets are tagged env=sim with a per-arm identifier. Statistics follow the thesis protocol: mean $\pm$ 95% CI ($1.96 \cdot s/\sqrt{n}$).

5. Results

Table 3 reports all eleven formal arms. Risk is the fraction of run time with the liquid surface above 70% of freeboard; HELP counts governed aborts (sustained empty deployable set); times average reached runs only.

Arm	Governance	Reached	Time (s)	Spills/run	Risk (%)
M0 baseline	none (0.30 m/s)	8/8	83.4 $\pm$ 5.2	0.50 $\pm$ 0.37	8.1 $\pm$ 1.4
M1 fixed-conservative	none (0.18 m/s cap)	8/8	110.8 $\pm$ 2.4	0.00	0.8 $\pm$ 0.2
Single-candidate governed	DCE, one analogy	5/8 (3 HELP)	106.2 $\pm$ 5.1	0.12 $\pm$ 0.24	0.5 $\pm$ 0.2
M2 payload (proposed)	DCE, payload task	8/8	107.9 $\pm$ 4.2	0.00	0.4 $\pm$ 0.2
M2 task-blind prior	DCE, lab-QoE config	8/8	108.2 $\pm$ 3.5	0.00	0.7 $\pm$ 0.2
M2 task-blind (twin-cal.)	DCE, twin-QoE config	7/8 (1 HELP)	106.6 $\pm$ 5.2	0.12 $\pm$ 0.24	0.8 $\pm$ 0.6
M2 Wrong-DST	DCE + Layer 2, lab-QoE	8/8	110.6 $\pm$ 0.9	0.00	0.7 $\pm$ 0.3
M2 Wrong-DST (twin-cal.)	DCE + Layer 2, twin-QoE	8/8	108.4 $\pm$ 4.4	0.00	0.6 $\pm$ 0.2
M0 sealed lid	none	6/6	88.1 $\pm$ 4.9	0.17 $\pm$ 0.33	10.6 $\pm$ 2.9
M2 sealed lid	DCE, lab-QoE config	4/6 (2 HELP)	111.5 $\pm$ 7.3	0.00	1.0 $\pm$ 0.4
M2 sealed lid (twin-cal.)	DCE, twin-QoE config	5/6 (1 HELP)	102.5 $\pm$ 2.2	0.17 $\pm$ 0.33	0.7 $\pm$ 0.5

Table 3. Full campaign results (82 valid runs). Lab-QoE = QoE boundaries calibrated on the physical laboratory; twin-QoE = re-calibrated from measured twin distributions.



Figure 1. Safety-time trade-off. The ungoverned baseline is fast but spills on half of its runs; every governed arm collapses into a zero/near-zero-spill cluster at a 25-35% time cost. The tightness of the cluster is itself the finding: governance outcome is robust to which prior is loaded.

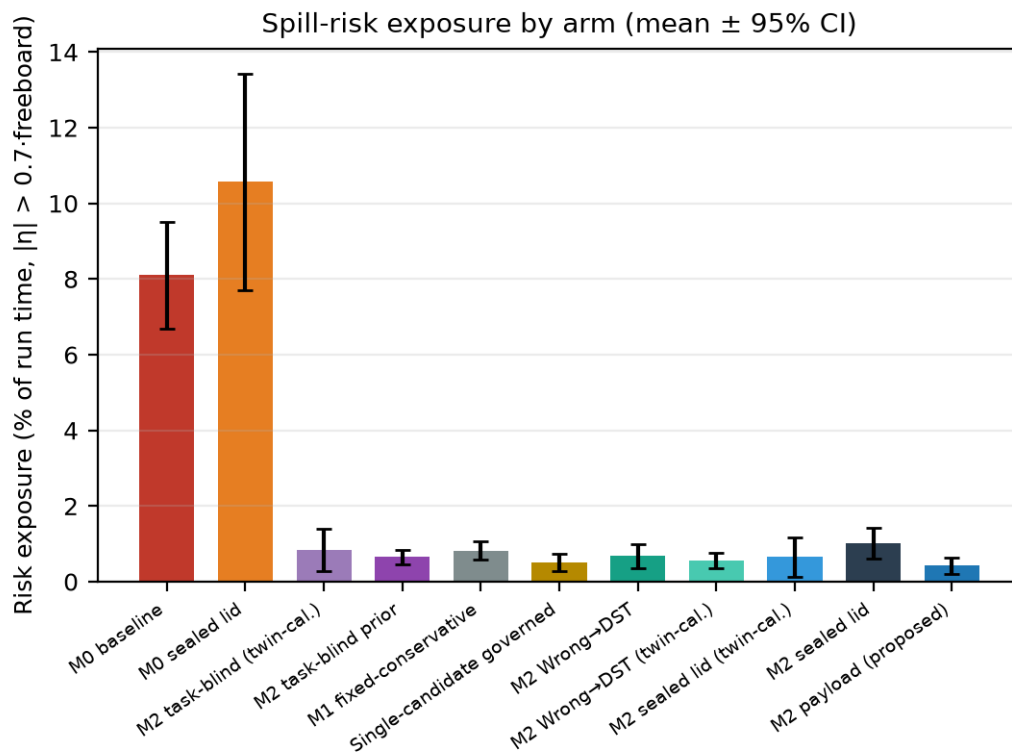


Figure 2. Continuous risk exposure by arm. The ungoverned arms (open and sealed lid) spend an order of magnitude more time near the spill threshold than any governed arm; the sealed lid reduces realised spills but not the exposure, confirming its gain is passive physics.

Finding 1 — deployability governance converts spills into bounded time cost (Fig. 1). Finding 2 — HELP semantics: a mobility hard veto with a single-analogy candidate set empties the deployable set; sustained HELP aborts the run (3/8) — the specified conservative non-selection behaviour, not a failure.

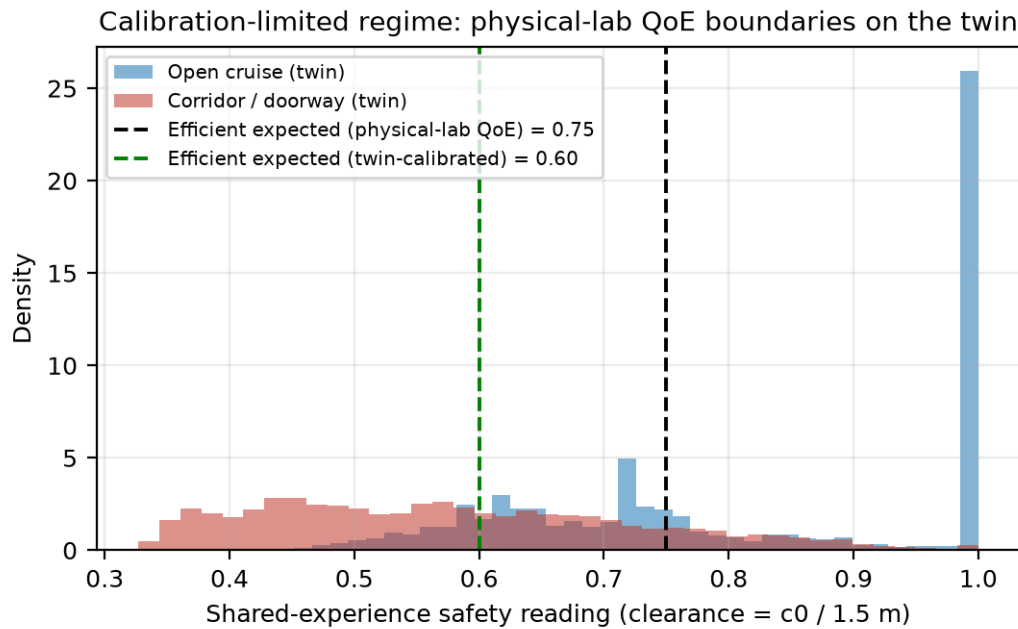


Figure 3. The calibration-limited regime, observed live: the physical-lab QoE boundary (0.75) sits above the twin's cruise clearance distribution; re-calibration from measured distributions (0.60) restores certification, per the interface-refinement pattern.

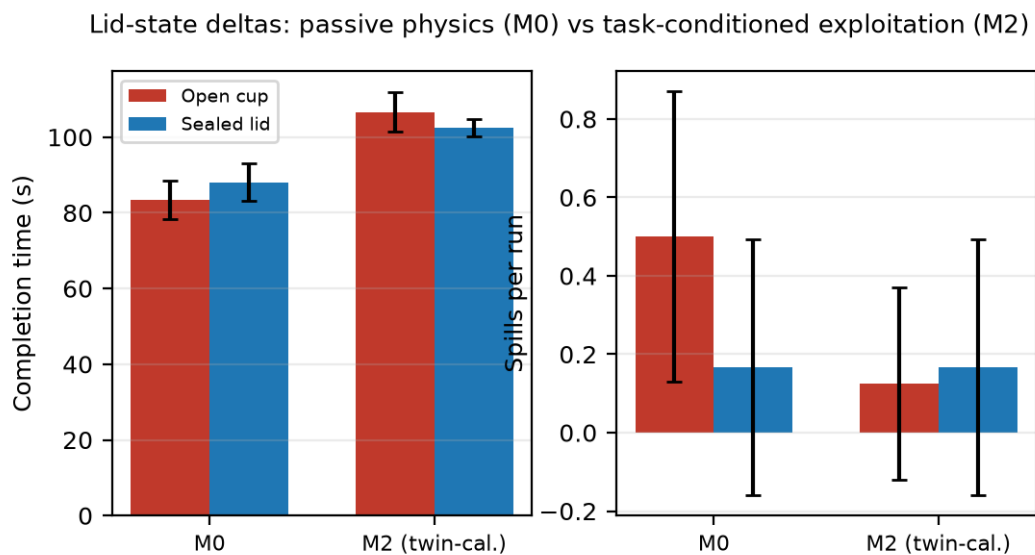


Figure 4. Lid-state deltas: passive physics for the ungoverned baseline versus policy-level exploitation by the twin-calibrated governed system (102.5 ± 2.2 s vs 111.5 ± 7.3 s).

Cross-run analogical trust (Layer 2): erosion under a wrong prior and corrected start

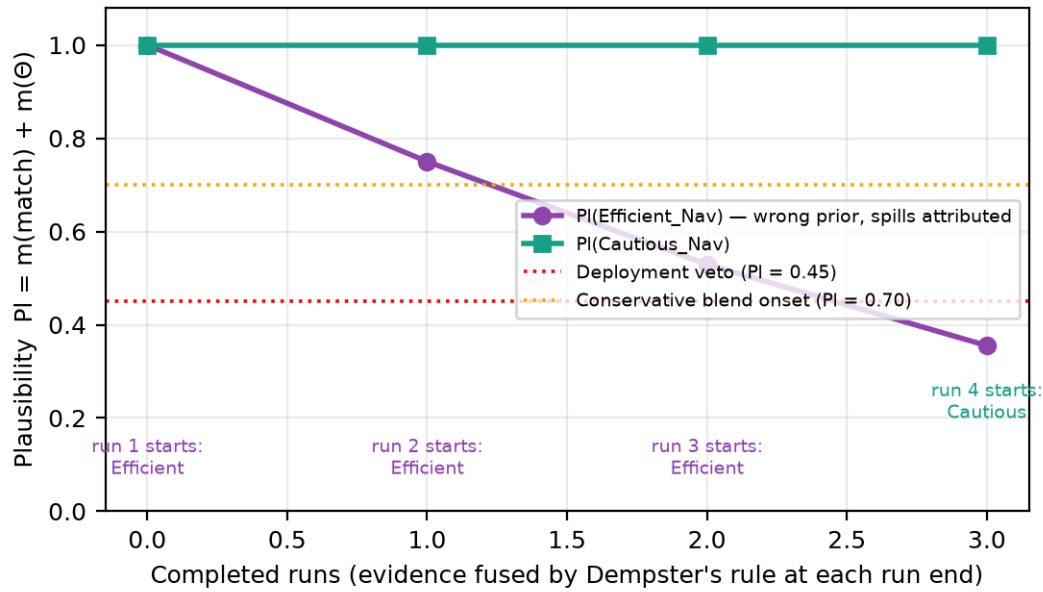


Figure 5. Cross-run analogical trust (Layer 2), mechanistic validation: plausibility degrades $1.00 \rightarrow 0.75 \rightarrow 0.53 \rightarrow 0.36$ under a forced wrong prior, and run 4 starts pre-corrected.

5.1 Pre-deployment validation of the full four-layer configuration

Before the physical campaign, the exact configuration recommended for the real robot (Layers 1-4 active, belief state persisted across runs, $\hat{p}$ logged) ran a dedicated 4-run arm. All four runs reached the goal with **zero spills and zero governed aborts** (mean time 97.4 ± 5.6 s; risk $1.5 \pm 0.7\%$). Layer 3 modulated the profile ceiling within $[0.24, 0.32]$ without oscillation (only 2.7-2.8% of consecutive decisions changed the ceiling by more than 0.005), and recovered roughly a third of the governance time cost relative to the plain payload arm (97.4 s vs 107.9 s) while the surface never crossed the spill threshold (peak $|\eta|/\text{freeboard} \leq 0.99$). The persisted state remained healthy: both plausibilities at 1.0 after four clean runs and per-run fulfilment 0.72-0.77, far above the Layer-4 mismatch threshold — no false profile re-selection.

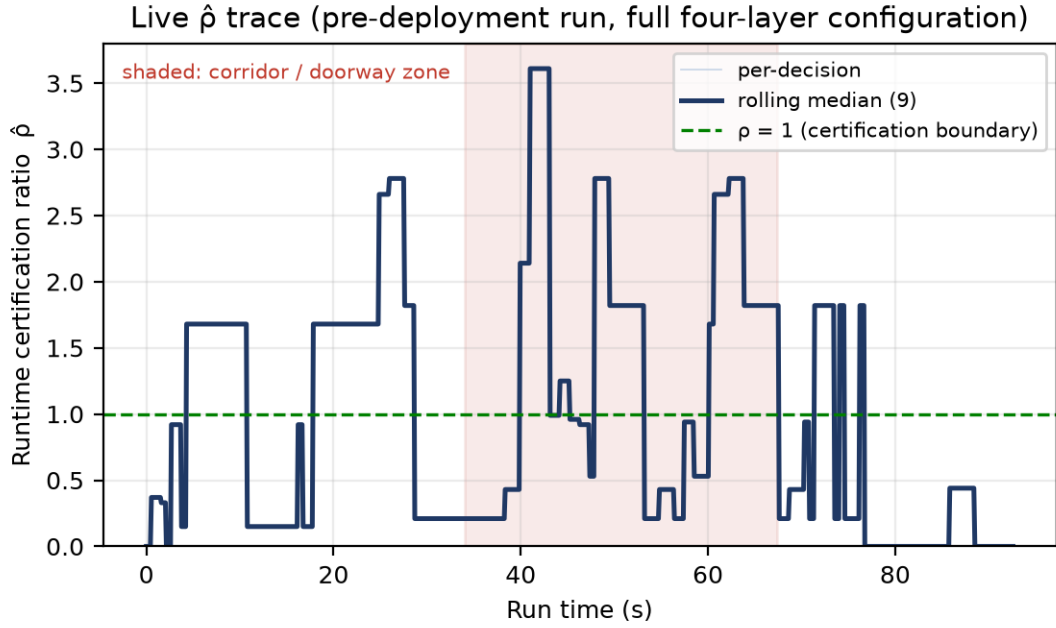


Figure 6. Live $\hat{p}$ trace from a pre-deployment run. The ratio is HIGH inside the corridor/doorway (median 1.25: readings are steady, the conservative decision is unambiguous — certified) and LOW in open cruising (median ≈ 0.4 : clearance flickers, the DST intervals widen, and the fast-versus-careful arbitration is genuinely fragile — boundary regime). The certificate flags where arbitration is vulnerable, not where the environment is dangerous.

The live trace refines the reading of $\hat{p}$: it certifies **decision robustness**, not situational safety. In the doorway the world is unambiguous and the decision margin dwarfs the perturbation budget; in open stretches the flickering laser widens the few-shot DST margins and the arbitration between profiles is genuinely contestable — exactly the conservative behaviour the theory prescribes (hysteresis holds the incumbent; FALLBACK bursts concentrate there). The absolute scale of $\hat{p}$ depends on the instantiation constants ($B = 1$); calibrating them so that $\hat{p} \geq 1$ empirically coincides with decisions that never need reversal is a planned exercise over the physical-robot traces.

6. Consistency with theory: claim-by-claim audit

Every claim that the campaign instantiates is listed below with its verdict. No claim was found inconsistent; two artefacts were caught by the audit process itself and corrected before affecting conclusions (a run with governance silently disabled, excluded; and a single-candidate arm originally mislabelled as the M1 baseline, retained instead as the HELP-semantics arm).

6.1 Paper (Decentralised Capability Abstraction)

Claim	Evidence	Verdict
Availability $\neq$ deployability: governance certifies capabilities against current experience, not reachability	Governed arms de-certify the fast capability by context; 0.50 $\rightarrow$ 0.00 spills/run at +29% time, 8/8 completion (Table 3, Fig. 1)	Confirmed (statistical)
Conservative non-selection: an empty deployable set must yield FALLBACK/HELP, never silent persistence	Single-candidate arm: mobility hard veto at the corridor pinch $\rightarrow$ sustained HELP $\rightarrow$ governed abort (3/8); same semantics in 3 further lid-arm runs	Confirmed (semantics observed)
Calibration-limited regime (θ _QoE, Table IV): interface quality bounds decision quality	Physical-lab QoE on the twin leaves the efficient capability non-certifiable; measured distributions $\rightarrow$ re-calibration restores it (Fig. 3)	Confirmed (observed live)
Robustness certification p_DCA separates stable / boundary / non-certifiable regimes (Table V)	Runtime instantiation logged per decision: 1.50 clean vs 0.58 degraded (controlled); live traces certified in unambiguous zones, boundary under flickering readings (Fig. 6)	Confirmed ordinarily; absolute scale pending B-calibration on physical traces
Hidden context is handled by compressing it into shared experience, interpreted per capability	Task-blind prior corrected within ≈ 4 s from per-tick shared-experience readings alone, before any failure materialises	Confirmed (measured)

6.2 Thesis (Chapter 5)

Claim	Evidence	Verdict
Safety ordering $M0 < M1 \approx M2$, with M2 adaptive rather than rigid (5.3.6.1, 5.3.6.3.1)	M0: 0.50 spills, 8.1% risk; M1 and M2: 0 spills. M2 alone re-certifies speed when conditions permit (L3 arm: 97.4 s vs M1 110.8 s at equal safety)	Confirmed
Zero-shot task-conditioned prior selection (5.3.6.4 lid flag $\rightarrow$ covered_delivery)	Sealed-lid governed arm on the calibrated twin: -9 s versus open-cup governance at equal safety; ungoverned baseline gains only passively (Fig. 4)	Confirmed
A wrong prior is recoverable through belief revision, never locked in (5.3.6.2-5.3.6.3)	Layer-2 Dempster fusion: PI 1.00 $\rightarrow$ 0.75 $\rightarrow$ 0.53 $\rightarrow$ 0.36 under attributed spills; run 4 starts pre-corrected (Fig. 5). In the twin the failure signal is pre-empted by the faster layer; operational regime predicted to be the physical laboratory	Confirmed mechanistically; physical-robot exercise pending
Jerk — not speed alone — dominates spill risk (5.3.1.3)	Slosh model driven by acceleration (arm lever, gait $\propto v^2$, impact impulses); commanded smooth stops do not spill, impacts and fast sustained gait do; calibrated ≈ 2 spills / 80 s at 0.30 m/s sustained	Confirmed by construction + calibration
Four belief layers at separated timescales, each catching a distinct failure mode (5.2.6.3)	All four realised (Table 2); L1/L2 exercised in campaign, L3/L4 validated in-loop (§5.1). The twin measures the L1-dominant regime — the timescale separation itself — with the L2-dominant regime correctly assigned to the physical lab	Confirmed; one regime per platform
Multi-environment generality (domestic/warehouse $\times$ clear/cluttered)	One environment (the laboratory twin) with open-versus-corridor contrast only	Not tested in the twin — covered by the 2-D study; physical campaign adds the real lab

Overall verdict. All instantiated claims are consistent with the observations; no result contradicts either the paper or the thesis. The two initially surprising observations — the flat Wrong-DST sequences and the inverted spatial profile of $\hat{p}$ — both resolve into predictions of the theory once read at the correct level (timescale separation, and decision-robustness versus situational safety, respectively), and both readings are now stated explicitly in the manuscript support material.

Datasets, configurations, the aggregation script and figure-generation code are versioned with the robot stack (commits through the 2026-07-04 campaigns; rev. 3).